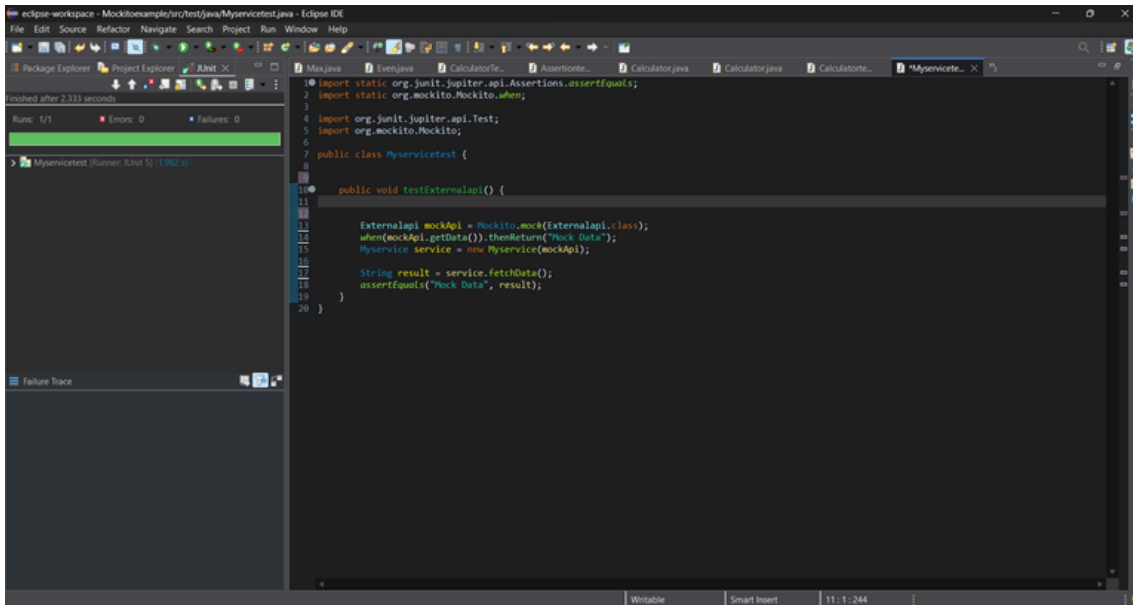


WEEK 1 ASSIGNMENT

Mockito Hands-On Exercises

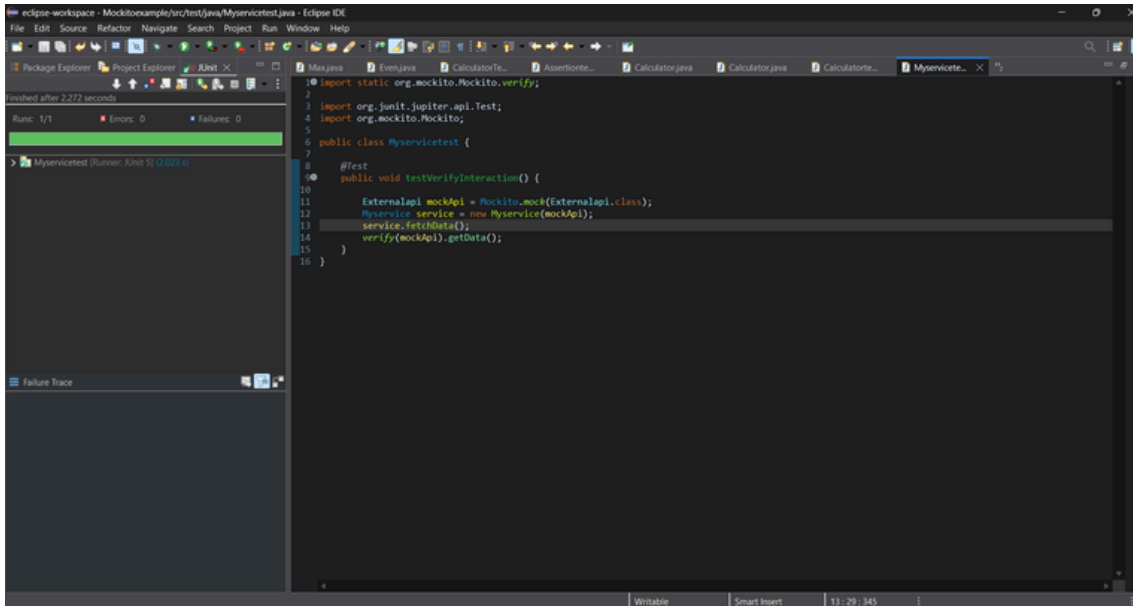
Exercise 1: Mocking and Stubbing



The screenshot shows the Eclipse IDE with a Java project named 'MockitoSample'. The 'MyServiceTest.java' file is open, showing a test method 'testExternalApi()' that uses Mockito to mock the 'ExternalApi' interface. The test method stubs the 'getData()' method to return 'Mock Data' and then calls 'fetchData()' on the 'MyService' object, which internally uses the 'ExternalApi' mock. The test passes, as indicated by the green progress bar in the Run console.

```
10 import static org.junit.jupiter.api.Assertions.assertEquals;
11 import static org.mockito.Mockito.when;
12
13 import org.junit.jupiter.api.Test;
14 import org.mockito.Mockito;
15
16 public class MyServiceTest {
17
18     @Test
19     public void testExternalApi() {
20
21         ExternalApi mockApi = Mockito.mock(ExternalApi.class);
22         when(mockApi.getData()).thenReturn("Mock Data");
23         MyService service = new MyService(mockApi);
24
25         String result = service.fetchData();
26         assertEquals("Mock Data", result);
27     }
28 }
```

Exercise 2: Verifying Interactions



The screenshot shows the Eclipse IDE with the same 'MockitoSample' project. The 'MyServiceTest.java' file is open, showing a test method 'testVerifyInteraction()' that uses Mockito to mock the 'ExternalApi' interface. The test method calls 'fetchData()' on the 'MyService' object, which internally uses the 'ExternalApi' mock. The test verifies that the 'getData()' method was called on the mock. The test passes, as indicated by the green progress bar in the Run console.

```
10 import static org.mockito.Mockito.verify;
11
12 import org.junit.jupiter.api.Test;
13 import org.mockito.Mockito;
14
15 public class MyServiceTest {
16
17     @Test
18     public void testVerifyInteraction() {
19
20         ExternalApi mockApi = Mockito.mock(ExternalApi.class);
21         MyService service = new MyService(mockApi);
22
23         service.fetchData();
24         verify(mockApi).getData();
25     }
26 }
```